

Conditional statements



1 Determine whether or not each of the following is a conjecture.

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| a Good. | b He is never going to get here. |
| c When does the next season come out? | d That bridge over there. |
| e The negative of x . | f How do we prove this? |
| g Dividing by zero is undefined. | h Axes. |
| i Colorless green ideas furious. | j What is your favourite color? |
| k Spider silk is stronger than steel. | l Don't sit too close to the screen. |
| m Welcome to all the new residents. | n Apples are made out of sunshine. |
| o Curved straight triangular. | p Corresponding angles are congruent. |
| q A triangle sometimes has five sides. | r Mark the point of intersection. |
| s Assume this is true for now. | t Who invented geometry? |
| u What does this button do? | v Yesterday when it all happened. |
| w The music is loud and the room is crowded. | x The moon orbits the earth. |

2 Determine whether or not each of the following is a conditional statement.

- a The moon orbits the earth.
- b The music is loud and the room is crowded.
- c If he replies to me, then we won't be late.
- d You can always draw a line parallel to another line.
- e The line intersecting two parallel lines.
- f The multiplication of two numbers.
- g Only equilateral triangles have three sides of equal length.

3 For the following conditional statements, write an equivalent statement.

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|-----------------------------------|----------------------------|
| a Parallel lines never intersect. | b No cat can speak French. |
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4 Determine whether or not each of the following statements is a conclusion.

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|---------------------|---------------------|
| a We won't be late. | b He replies to me. |
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- 5 Write the negation of the following statements:
- a All birds are able to fly.
 - b Some turtles do not have claws.
 - c No Libras are Geminis.
 - d No bicycles have four wheels.
 - e All houses are connected to the internet.
- 6 Consider the conditional statement "If C then D ", where C and D are both conjectures:
- a Write the converse of the statement.
 - b Write the inverse of the statement.
 - c Write the contrapositive of the statement.
- 7 Write the converse of the following statements:
- a If an animal is a bear, then it has claws.
 - b If a boomlug is round, then it is delicious.
 - c If a figure is a square, then it is a quadrilateral.
- 8 Write the contrapositive of the following statements:
- a If an animal is a bird, then it has feathers.
 - b If a number ends in zero, then it is divisible by ten.
 - c If a bamboozle is square, then it is loud.
- 9 Write the inverse of the following statements:
- a If an animal is a bird, then it has feathers.
 - b If two angles form a right angle, then they are complementary.
 - c If a heffelump is lost in the woods, then it is a woozle.
- 10 Complete the statement below.
The hypothesis of a conditional is the conclusion of its $\neg$.
- converse
 - inverse
 - negation
 - contrapositive

Additional questions



- 11** Write a conditional statement using a hypothesis p and conclusion q .
- 12** Does a conditional statement need to be true?
- 13** Identify which of the following are conditional statements.
- a** If a polygon has three sides, then it is a triangle.
 - b** The church is on Delaney Avenue.
 - c** If I go to school in Florida, then I live in Orlando.
 - d** If it is raining, then we cannot play baseball.
 - e** If a parallelogram is not a rhombus, then it is not a square.
 - f** Red and green are composite colors.
- 14** For each of the following conditional statements, identify the hypothesis and conclusion.
- a** If two angles are vertical, then they are congruent.
 - b** If the sum of a polygon's interior angle measures is 180° , then the polygon is a triangle.
 - c** Two triangles are congruent if they have the same side lengths.
 - d** When two lines are parallel, the consecutive interior angles formed by a transversal are supplementary.
- 15** Consider the conditional statement "If I can paint a picture, then I am an artist". Write the:
- a** Converse statement
 - b** Inverse statement
 - c** Contrapositive statement
- 16** For each pair of conditional statements, determine if they are converse, inverse or contrapositive statements.
- a** If two numbers are even, then their sum will be even.
If the sum of two numbers is even, then the two numbers are even.
 - b** If two lines do not meet at a right angle, then they are not perpendicular.
If two lines are perpendicular, then they meet at a right angle.
 - c** If two triangles are similar, they must have three congruent angles.
If two triangles are not similar, they cannot have three congruent angles.



17 Determine if each conditional statement is true or false. If it is false, give a counterexample.

- a** If I need to go shopping, then I will go to Walmart.
- b** If I have a part-time job, then I am part of the workforce.
- c** If I score a run in baseball, then I have touched all four bases.
- d** If I go to school, then I am a student.

18 Consider the statement "If $x = 2$, then $x^2 = 4$ ".

- a** Is this statement true? If not, explain why.
- b** Is the converse of this statement true? If not, explain why.
- c** Is the inverse of this statement true? If not, explain why.
- d** Is the contrapositive of this statement true? If not, explain why.

19 Write a conditional statement which is true and its converse is not true. Give an example for why the converse is not true.

20 Consider the statement for the vertical angles theorem:
"Vertical angles are congruent."

- a** Write the theorem as an "if...then" statement.
- b** Write a conditional statement for the contrapositive of the vertical angles theorem.

21 Rewrite the following as conditional statements.

- a** You miss 100% of the shots you don't take.
- b** The bigger they are, the harder they fall.
- c** When in doubt, tell the truth.